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Title: What Makes a Steam Game Successful? A Statistical Analysis of performance of Paid Steam Games

Abstract

The commercial drivers of PC game success are of substantial interest to developers and publishers, but the relative importance of critic reviews, user reviews, price, genre, playtime, and multiplayer features is not well established. Using a March 2025 snapshot of the Steam catalog from Kaggle ($n = 1,557$ paid games with Metacritic scores and non-zero playtime), we construct a composite success metric defined as the product of total review count and price, and examine seven research questions via simple and multiple linear regression, a Welch two-sample t-test, and Spearman correlation.

Metacritic critic score was a stronger single predictor of log success than Steam user-review percentage (adjusted $R^2 = 0.204$ vs. 0.137), and combining both yielded a modest improvement (adjusted $R^2 = 0.222$). After controlling for overlapping genre tags, Simulation, Action, and RPG titles were associated with higher success, while the Indie tag carried a substantial negative penalty. Price was positively correlated with success, multiplayer games outperformed single-player titles, and price efficiency showed essentially no monotonic relationship with success. Findings are limited to paid, critic-reviewed Steam releases.

1. Introduction

The video game industry generated over \$180 billion in revenue in 2024, with PC games, and specifically titles distributed through Valve's Steam platform, accounting for a substantial share. For independent developers and large studios alike, the question of what drives a game's commercial performance on Steam is both economically consequential and methodologically interesting. Unlike many consumer products, games are subject to a tangle of overlapping signals of quality including professional critic reviews aggregated by sites like Metacritic, user-generated review percentages on the Steam storefront itself, genre conventions, pricing strategy, and player engagement as measured by hours played.

This paper takes a regression-based approach to disentangling these factors. The data are a March 2025 snapshot of the Steam catalog containing rich per-game information on prices, reviews, playtimes, genres, and categories. Because reviews and prices alone do not capture commercial outcomes, (a \$60 AAA title with 10,000 reviews represents a very different business outcome from a \$5 indie with 10,000 reviews) we construct a composite success metric that weights review volume by price, on the reasoning that each review is a strong proxy for a sale and that revenue scales with price.

We address seven research questions:

Q1. Is the Metacritic critic score or the Steam user-review percentage a better single predictor of success?

Q2. Does combining both scores meaningfully improve prediction over the better single predictor?

Q3. Are some genres systematically associated with higher success after controlling for other genre tags?

Q4. Is higher price positively correlated with success?

Q5. Is median playtime or average playtime the better predictor of success?

Q6. Do multiplayer games achieve higher success than single-player games?

Q7. Is price efficiency (median hours per dollar) associated with success?

Our prior expectations, based on common Steam-developer folklore, were: user scores beat critic scores; combining them helps; certain genres like action dominate; higher price is positively correlated with success because of correlation with budget; median playtime is more reliable than average because of long-tail super-fans; single-player games outperform multiplayer titles; and price efficiency rewards developers who deliver more hours per dollar.

2. Method

2.1 Data

Our dataset is called “Steam Games Dataset 2025”, which we found on [Kaggle](#). Each row in the dataset represents one game and contains information such as name of the game, description of the game, critic score, price, playtime, reviews, etc. For our

analysis, we are using variables such as `metacritic_score`, `price`, `median_playtime_forever`, `average_playtime_forever`, `pct_pos_total`, `num_reviews_total`, `genres`, and `categories`.

Variable `metacritic_score` represents the critic score of each game, while `price` shows the cost of the game. Variable `median_playtime_forever` represents the median number of minutes players have spent on the game and variable `average_playtime_forever` measures the average number of minutes played. Variable `pct_pos_total` represents the percentage of positive reviews out of total reviews and variable `num_reviews_total` shows the total number of reviews a game has received. Also the variable `genres` is a string that lists the genres associated with each game while `categories` is a comma separated string that we use to determine whether a game is a multiplayer game.

For this analysis, we restricted the sample to games with a Metacritic score, a strictly positive price (because the success metric multiplies by price), and non-zero recorded median and average playtime. After filtering, $n = 1557$ games remained. This sub-sample skews toward higher-budget, critic-reviewed paid releases, a population for which the metrics of interest are meaningful, but findings should not be generalized to free-to-play or unrated indie titles.

2.2 Derived variables

We used `mutate()` functions to create new variables that help us answer our research question. First, we create the variable `success_score`, which was calculated as `num_reviews_total * price`. We also created the variable `price_efficiency`, which was calculated as `median_playtime_forever / price`. To examine the multiplayer features within the game, we create the variable `is_multiplayer`, which is marked as “True” if categories contain “Multi-player”. Based on this, we created another variable `player_mode`, which labels a game as “multiplayer” if `is_multiplayer` is true.

We noticed that some of the variables were right-skewed, so we used log transformations to make the residuals closer to normal. The variables that we used for log transformation were `log_success_score`, `log_median_playtime_forever`, `log_average_playtime_forever`, `log_price_efficiency`, and `log_num_reviews`. Each of these variables represents the logged version of its original variable. We also created 9 genre related variables to separate the different genres listed for each game, which allow us to analyze genre more clearly.

2.3 Analysis Plan

We will use the Steam Games Dataset 2025 from Kaggle to study which factors are related to the success of games. Each row in the dataset represents one game, and the dataset contains about 90,000 games. For this project, we define a game’s success score as the number of reviews multiplied by the price of the game, which gives an approximate measure of sales performance. All analyses were conducted in R using

tidyverse, broom, and base stats. For each research question we fitted one or more ordinary least squares regressions of log_success on the relevant predictor(s). To begin the analysis, we will clean the dataset by selecting the relevant variables, removing missing values when necessary, and creating new variables such as success score and price efficiency.

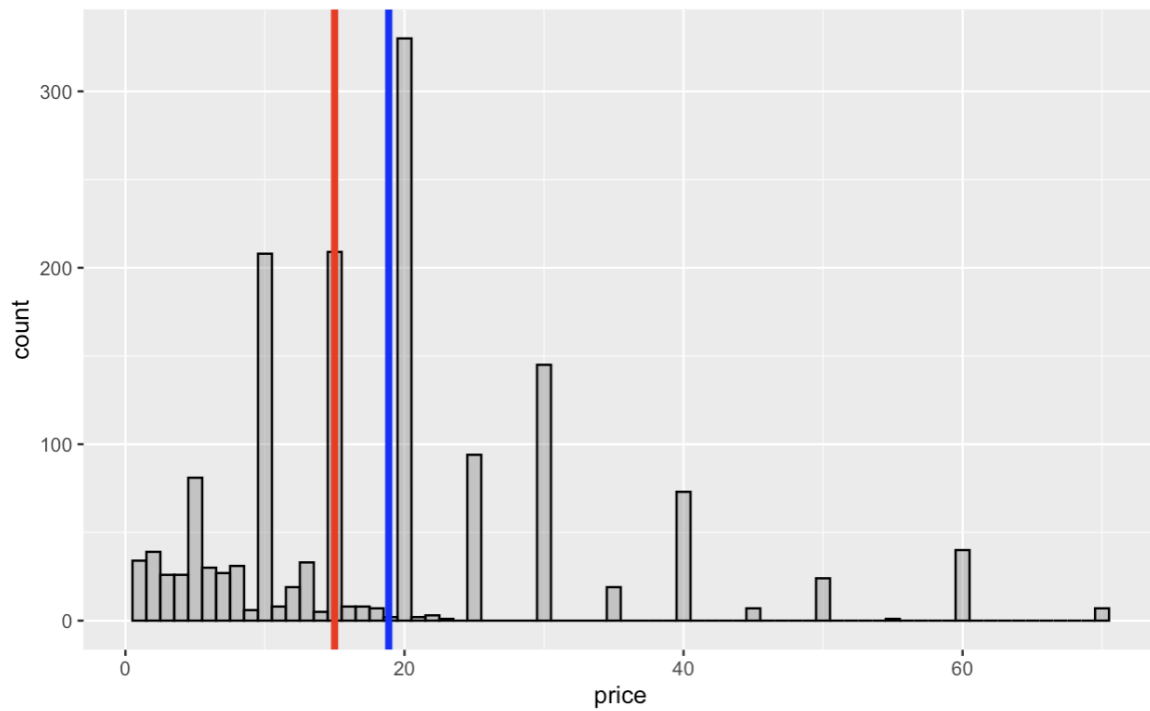
Log-transformation of the response was used as the primary remedy for the strong right-skew in success_score, and log-transformation of the playtime predictors and price_efficiency was used when they exhibited similar skew. Standard regression diagnostics (residuals-vs-fitted, normal Q-Q, scale-location, residuals-vs-leverage) were inspected for every fitted model to verify approximate linearity, constant variance of residuals, and roughly normal residuals. Log-transformation of the response was used as the primary remedy for the strong right-skew in success_score, and log-transformation of the playtime predictors and price_efficiency was used when they exhibited similar skew.

We will use scatterplots to analyze relationships between quantitative variables, such as price versus success, user score versus success, critic score versus success, and playtime versus success. We will use side-by-side boxplots to compare success across categorical variables, such as genre and single-player versus multiplayer games. To test our research questions, we will use simple linear regression to compare whether critic scores or user scores are better individual predictors of success, and multiple linear regression to determine whether combining both predictors creates a stronger model.

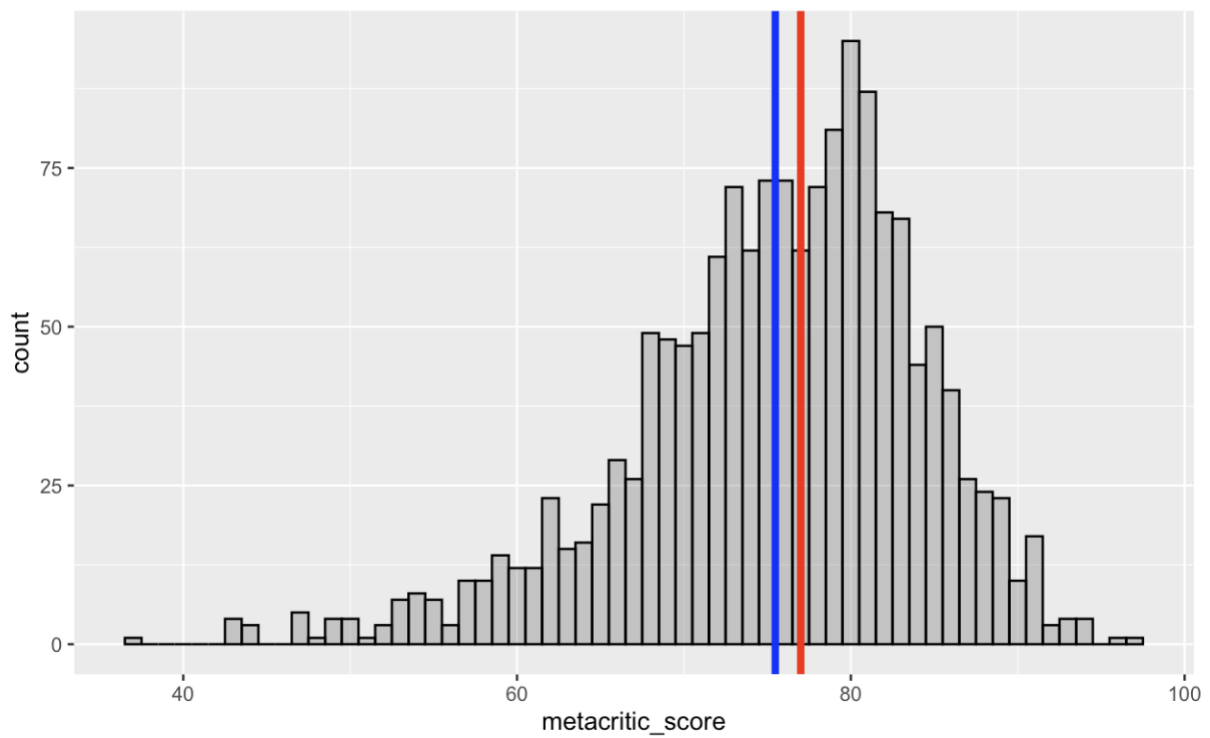
We will also use regression to evaluate whether price, median playtime, and average playtime, are correlated with success. To compare success between single-player and multiplayer games, we will use a two-sample t-test. To compare success across multiple genres, we will use multiple linear regression with nine binary indicator variables. For all regression models, we will check whether the analysis is appropriate by examining residual plots for linearity and constant variance, and QQ plots or histograms to check whether the residuals are approximately normal. We used spearman correlation to check the correlation between success score and price efficiency. These methods will allow us to evaluate which game characteristics are most strongly associated with success. For Q4 we supplemented the regression with a Pearson correlation between price and the untransformed success_score, since the relationship is the question and correlation provides a more interpretable effect size. Significance was assessed at $\alpha = 0.05$ throughout.

3. Results

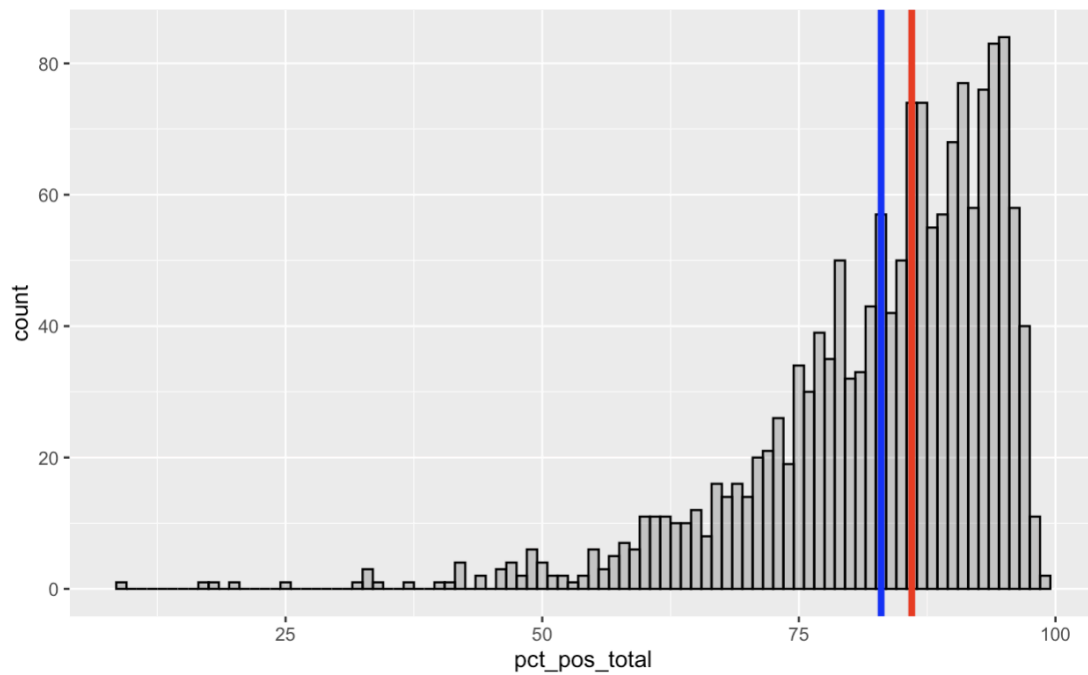
3.1 Sample Description



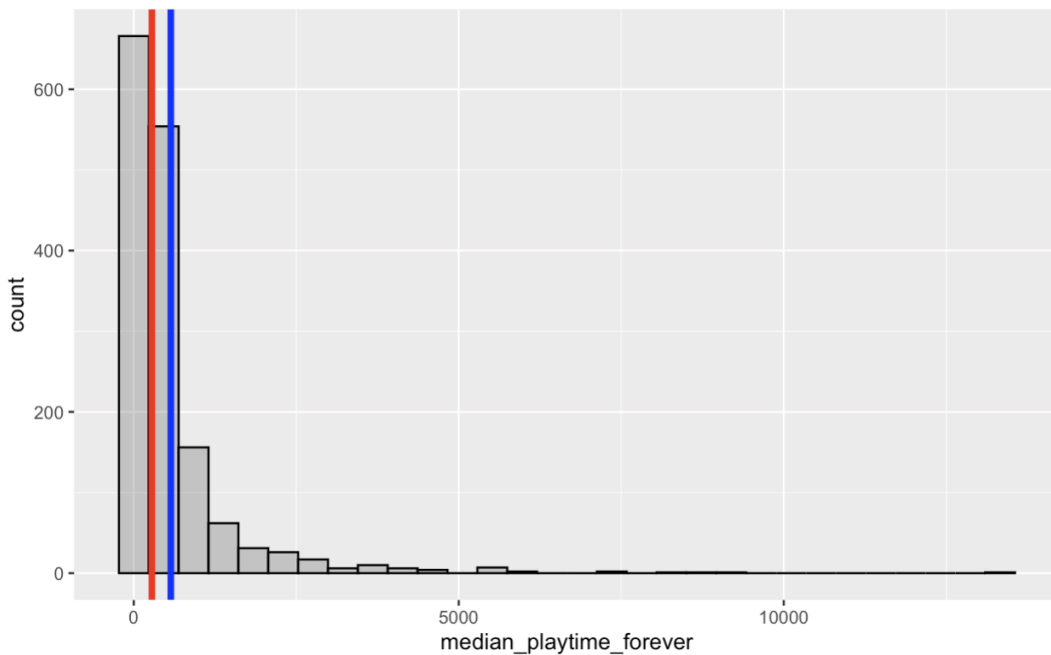
The analytic sample of 1,557 games has prices ranging from below \$1 to \$70, with \$20 being the most common price, and a long tail of higher-priced AAA titles.

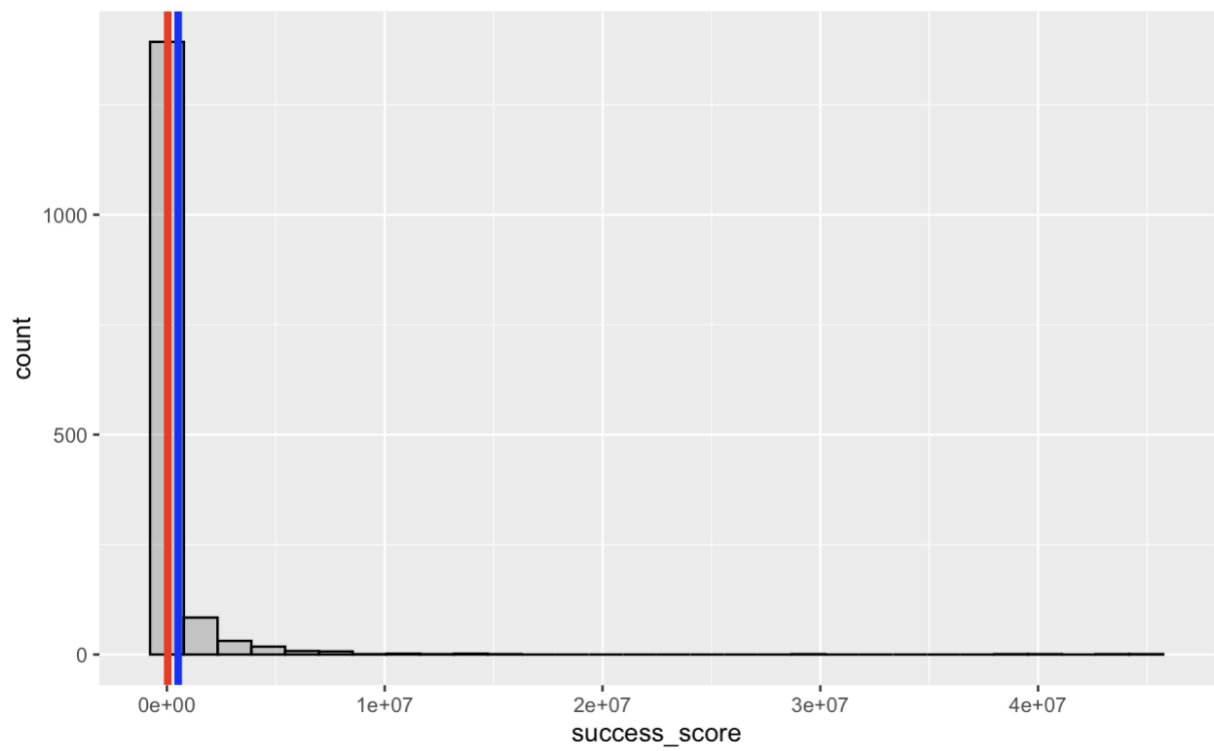
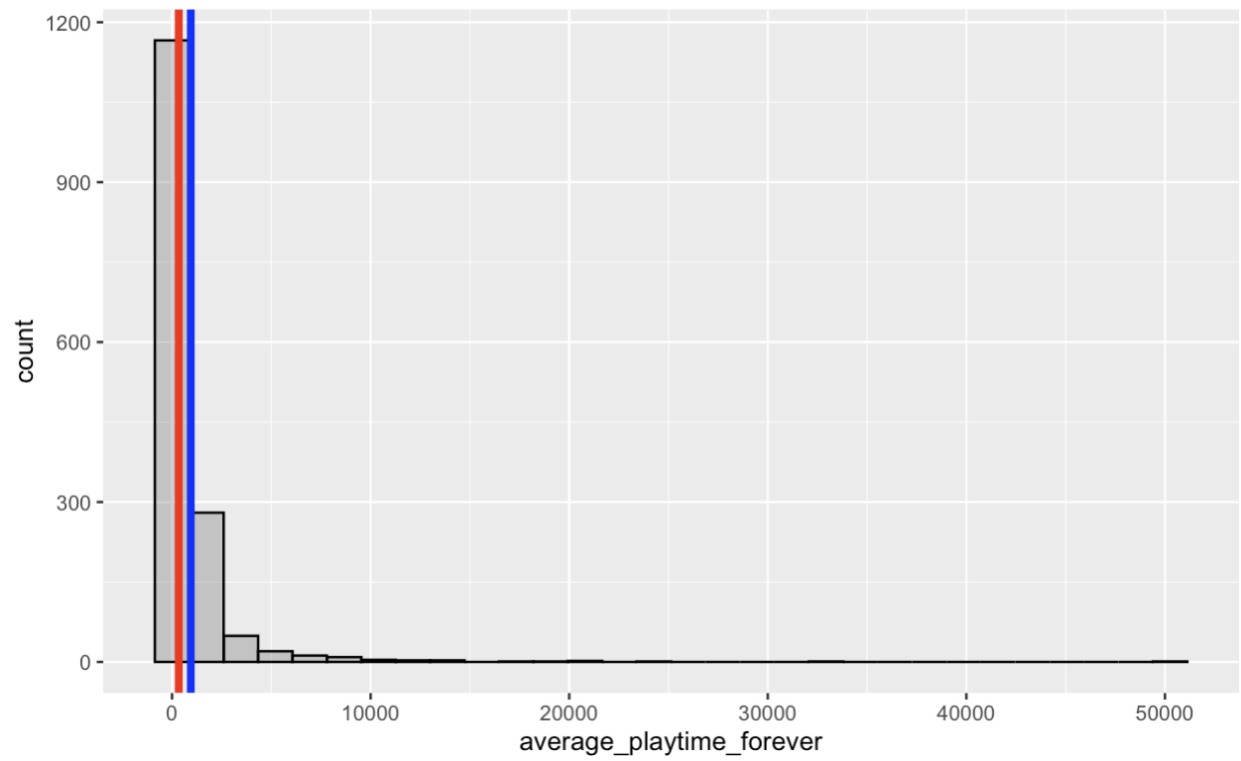


Metacritic scores cluster between the mid-60s and mid-80s, with very few games scoring below 50.



The Steam user-positive percentage is sharply left-skewed with 96% being the most positive score.





Both playtime variables and success_score itself are extreme right-skewed, motivating the log transformation used throughout.

Of the nine genre tags, Indie (934 games), Action (781), and Adventure (757) are by far the most common; Sports (39) and Racing (40) are sparse, and effect estimates for those should therefore be interpreted with caution. The sample contains 542 multiplayer-flagged games and 1,015 single-player-only games.

3.2 Question 1: Is the Metacritic critic score or the Steam user-review percentage a better single predictor of success?

Before comparing the models, we checked the diagnostic plots for the two single-predictor regression models. Overall, the plots were reasonably acceptable for using linear regression. In the Residuals vs Fitted plots, the residuals were mostly centered around 0, although there was some slight curvature, so the linearity condition was not perfect. The Q-Q plots were close to a straight line, suggesting that the residuals were approximately normal. The Scale-Location plots showed that the residual spread was mostly consistent, although there was some slight variation. The Residuals vs Leverage plots did not show any serious influential points based on Cook's distance. Therefore, it was reasonable to use linear regression to compare the two predictors, although the conditions were not perfect.

To answer this question, we fit two simple linear regression models using `log_success` as the response variable. The first model used `metacritic_score`, which represents Metacritic critic score, as the predictor. The second model used `pct_pos_total`, which represents Steam user-review percentage, as the predictor. Both single-predictor regression models were statistically significant, meaning that both Metacritic critic score and Steam user-review percentage have a statistically significant relationship with `log_success`. The model using `metacritic_score` had an adjusted R^2 of 0.204, while the model using `pct_pos_total` had an adjusted R^2 of 0.137.

Since the `metacritic_score` model has the higher adjusted R^2 , it explains more variation in `log_success` than the `pct_pos_total` model. Therefore, Metacritic critic score is a better single predictor of game success than Steam user-review percentage.

3.3 Question 2: Does combining both scores meaningfully improve prediction over the better single predictor?

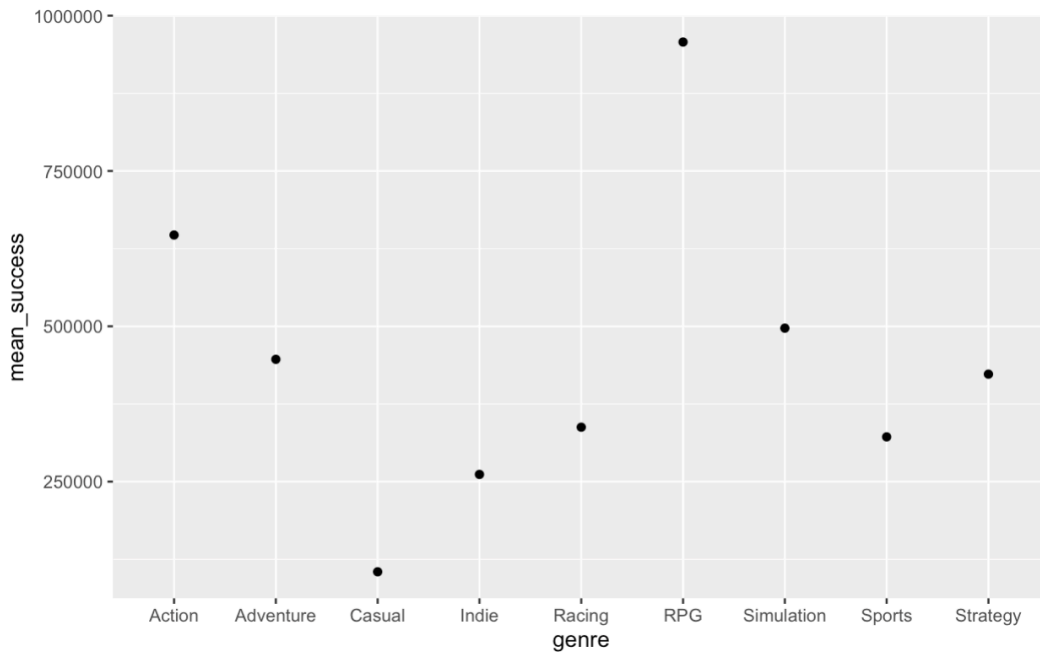
To answer this question, we fit a multiple linear regression model using both `metacritic_score` and `pct_pos_total` as predictors of `log_success`. We also checked the diagnostic plots for this combined model. The Residuals vs Fitted plot showed residuals mostly centered around 0, with a slight curved pattern. The Q-Q plot was close to a straight line, suggesting that the residuals were approximately normal. The Scale-Location plot showed mostly consistent residual spread, although there was still some variation. The Residuals vs Leverage plot did not show any serious influential

points based on Cook's distance. Overall, the combined model was reasonably appropriate to use, although the model conditions were not perfect.

Both predictor regression models were statistically significant, meaning that they both have a statistically significant relationship with `log_success`. From Question 1, the better one-predictor model was the Metacritic critic score model, with an adjusted R^2 of 0.204. The combined model using both Metacritic critic score and Steam user-review percentage had an adjusted R^2 of 0.222. Since the adjusted R^2 increased from 0.204 to 0.222, the combined model is stronger than the better one-predictor model.

However, the improvement is modest. The combined model explains about 22.2% of the variation in `log_success`, compared to about 20.4% for the Metacritic critic score model. Therefore, combining Metacritic critic score and Steam user-review percentage does improve prediction, but the improvement is relatively small rather than dramatic.

3.4 Question 3: Are some genres systematically associated with higher success after controlling for other genre tags?



genre	n	mean_success	median_success
<chr>	<int>	<dbl>	<dbl>
Action	781	644411.3	52904.88
Adventure	757	446245.7	34758.41
Casual	203	104860.2	14595.39
Indie	934	261044.8	23754.35
RPG	382	955037.5	57461.25
Simulation	286	496999.3	45299.78
Strategy	406	422948.8	36598.62
Sports	39	321980.1	14532.73
Racing	40	329143.9	26920.29

Mean success scores varied substantially by genre tag, with RPG titles averaging the highest unadjusted success (\$955037.5 mean), followed by Action (\$644411.3) and Simulation (\$496,999); Casual titles averaged the lowest (\$104,860). However, because genre tags overlap heavily, for example an RPG game is also often tagged Action and Indie, the relevant question is the partial effect of each tag after controlling for the others. We therefore fit a single regression of log_success on all nine flags simultaneously.

The model achieved adjusted $R^2 = 0.134$ with strongly significant coefficients for most tags:

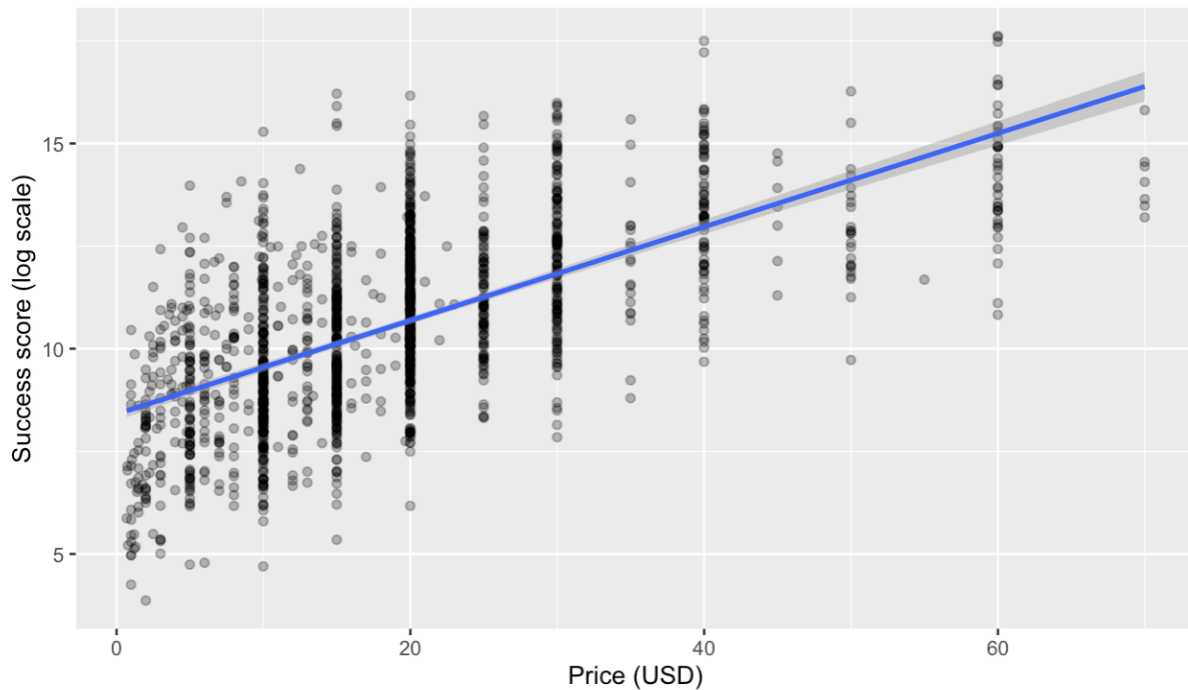
term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	10.6864691	0.1449588	73.7207276	0.000000e+00
is_actionTRUE	0.6998169	0.1167405	5.9946369	2.534701e-09
is_adventureTRUE	0.2672928	0.1205195	2.2178384	2.671103e-02
is_casualTRUE	-0.7031785	0.1670221	-4.2100924	2.700004e-05
is_indieTRUE	-1.2611269	0.1151051	-10.9563099	5.924738e-27
is_rpgTRUE	0.6514691	0.1286637	5.0633464	4.614492e-07
is_simulationTRUE	0.8194878	0.1526652	5.3678763	9.182410e-08
is_strategyTRUE	-0.1848326	0.1422417	-1.2994259	1.939919e-01

is_sportsTRUE	-0.4745764	0.4212965	-1.1264667	2.601432e-01
is_racingTRUE	-0.3419704	0.4213843	-0.8115404	4.171806e-01

Three findings stand out. First, the Indie tag carries an enormous negative penalty of -1.26 on the log scale, meaning that after controlling for other tags, indie-labeled games achieve only about 28% ($\exp(-1.26) \approx 0.28$) of the success score of otherwise comparable non-indie titles. This is striking but not surprising since the Indie label is essentially a self-identified marker of small-team, low-marketing production, and the `success_score` metric is sensitive to both review volume and price, both of which are systematically lower for indie titles. Second, Simulation, Action, and RPG emerge as the strongest positive predictors, each roughly doubling the expected success score. Simulation is somewhat unexpected given the relative niche-ness of the genre, but the result is consistent with the existence of long-tail simulation hits like *Cities: Skylines*, *Stardew Valley*, and the *Train/Truck Simulator* franchises. Third, the Strategy, Sports, and Racing coefficients are negative but not significant. This is almost certainly because the sports and racing tags contain only 39 games in the analytic sample, while the strategy tag has a coefficient that is too small in magnitude. Therefore, we decline to draw conclusions about those genres.

3.5 Question 4: Is higher price positively correlated with success?

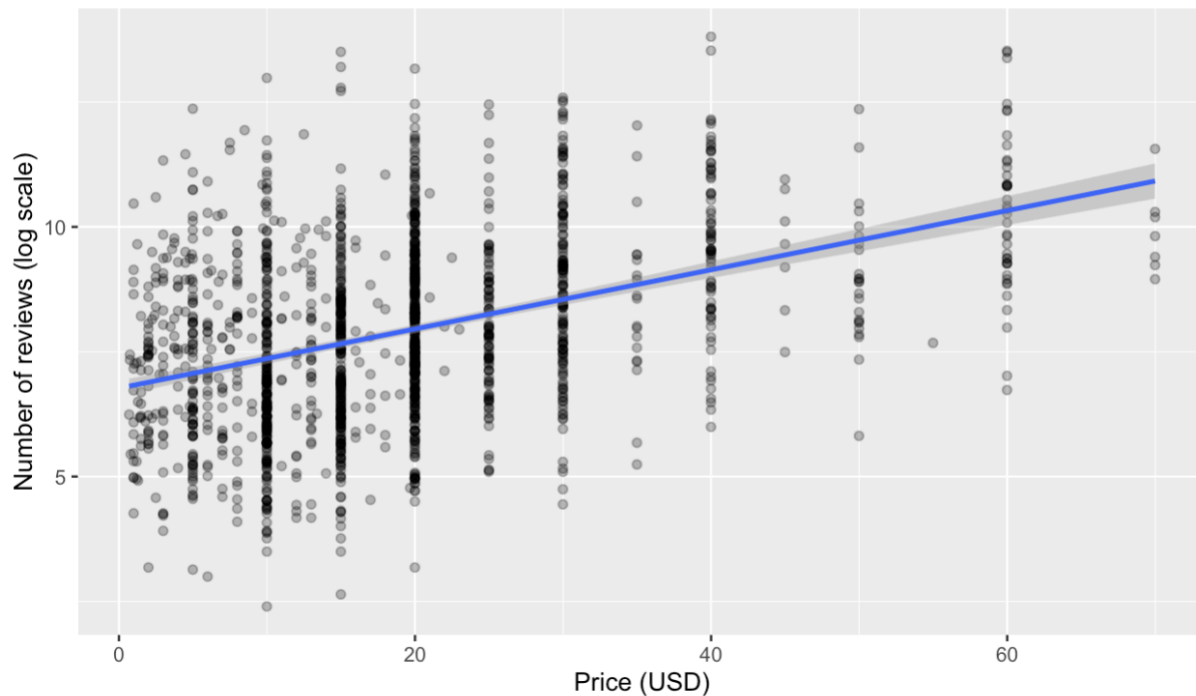
Q4: Price vs. log(success score)



Price was positively and significantly correlated with `log_success`. Pearson's $r = 0.639$, and in the LS regression of `log_success` on price, the slope was 0.114 per dollar ($SE = 0.003$, $p < 0.001$) with adjusted $R^2 = 0.408$. This means that a \$10 increase in price is associated with roughly a 200% ($\exp(1.14) - 1 \approx 2.127$) increase in `success_score`.

Because `success_score` = `num_reviews_total` $\times$ price, doubling price doubles `success_score` if review count is unchanged. Since price appears in `success_score` by construction, we also checked the relationship using just review count as the outcome.

Q4: Price vs. log(number of reviews)



Price was also positively and significantly correlated with `log_num_reviews`. Pearson's $r = 0.404$, and in the LS regression of `log_num_reviews` on price, the slope was 0.059 per dollar ($SE = 0.003$, $p < 0.001$) with adjusted $R^2 = 0.163$. This means that a \$10 increase in price is associated with roughly a 80% ($\exp(0.59) - 1 \approx 0.804$) increase in `num_reviews_total`. This means that higher-priced games attract proportionally more reviews. They sell more copies, not fewer. The conventional wisdom that high prices suppress audiences is not borne out here, at least within the range of \$1–\$70 represented in the data. We interpret this as a reflection of the underlying budget-quality correlation: more expensive games are typically AAA titles with marketing budgets, established franchises, and broader appeal, which drives both their price and their unit sales.

Question 5: Is median or average play time a better predictor of success?

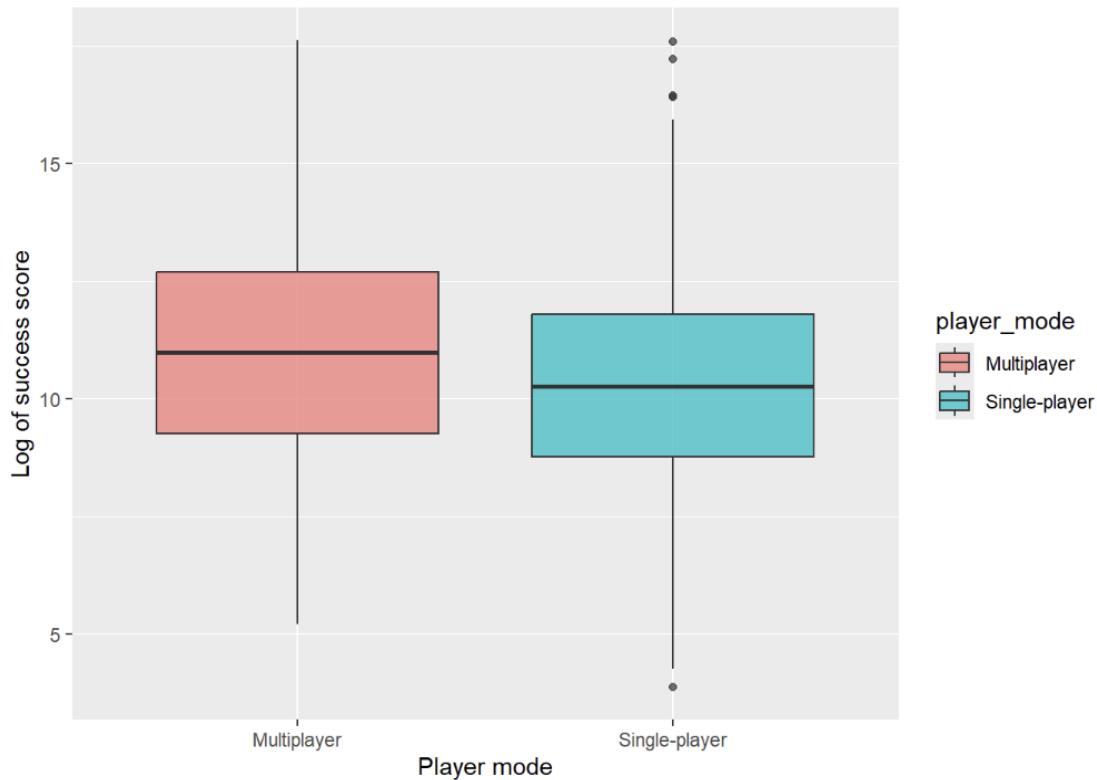
To answer this question, we fit two simple linear regression models using `log_success` as the response variable. The first model used `log_median_playtime_forever` as the predictor, and the second model used `log_average_playtime_forever` as the predictor. Before comparing the models, we checked the regression diagnostic plots for both models. For both models, the Q-Q plots are mostly close to a straight line, suggesting that the residuals are approximately normal. The residuals vs fitted plots show some slight curved patterns, so the linearity condition is not perfect, but the models are still reasonably usable for comparison. The scale-location plots show some increase in spread, meaning the constant variance condition is not perfect. The residuals vs leverage plots show a few points with higher leverage, but there do not appear to be extreme influential points that dominate the model. Overall, the diagnostic plots suggest that the models are not perfect, but they are acceptable for comparing the two predictors.

Both single-predictor regression models were statistically significant, meaning that both median play time and average play time have a statistically significant relationship with `log_success`. The adjusted R^2 for the model using `log_median_playtime_forever` is 0.2155, which means this model explains about 21.55% of the variation in `log_success`. The adjusted R^2 for the model using `log_average_playtime_forever` is 0.3785, which means this model explains about 37.85% of the variation in `log_success`.

Since the adjusted R^2 is much higher for the model using `log_average_playtime_forever`, average play time is the better predictor of success. In context, this means that the average amount of time players spend on a game has a stronger relationship with game success than the median amount of time players spend on a game. Therefore, based on these regression models, average play time is a better predictor of success than median play time.

3.7 Question 6: Do multiplayer games achieve higher success than single-player games?

For Q6, we examined whether multiplayer games achieve higher success than single-player games. Success was measured using the log of the success score, where the success score was calculated from the total number of reviews multiplied by price. After cleaning the data, there were 542 multiplayer games and 1015 single-player games included in the analysis.



The side-by-side boxplot shows that multiplayer games tend to have a higher log success score than single-player games. The median log success score for multiplayer games appears higher than the median for single-player games, suggesting that multiplayer games may be more successful on average.

The hypotheses are:

Null Hypothesis: Multiplayer games and single-player games have the same average log success score.

Alternative Hypothesis: Multiplayer games have a higher average log success score than single-player games.

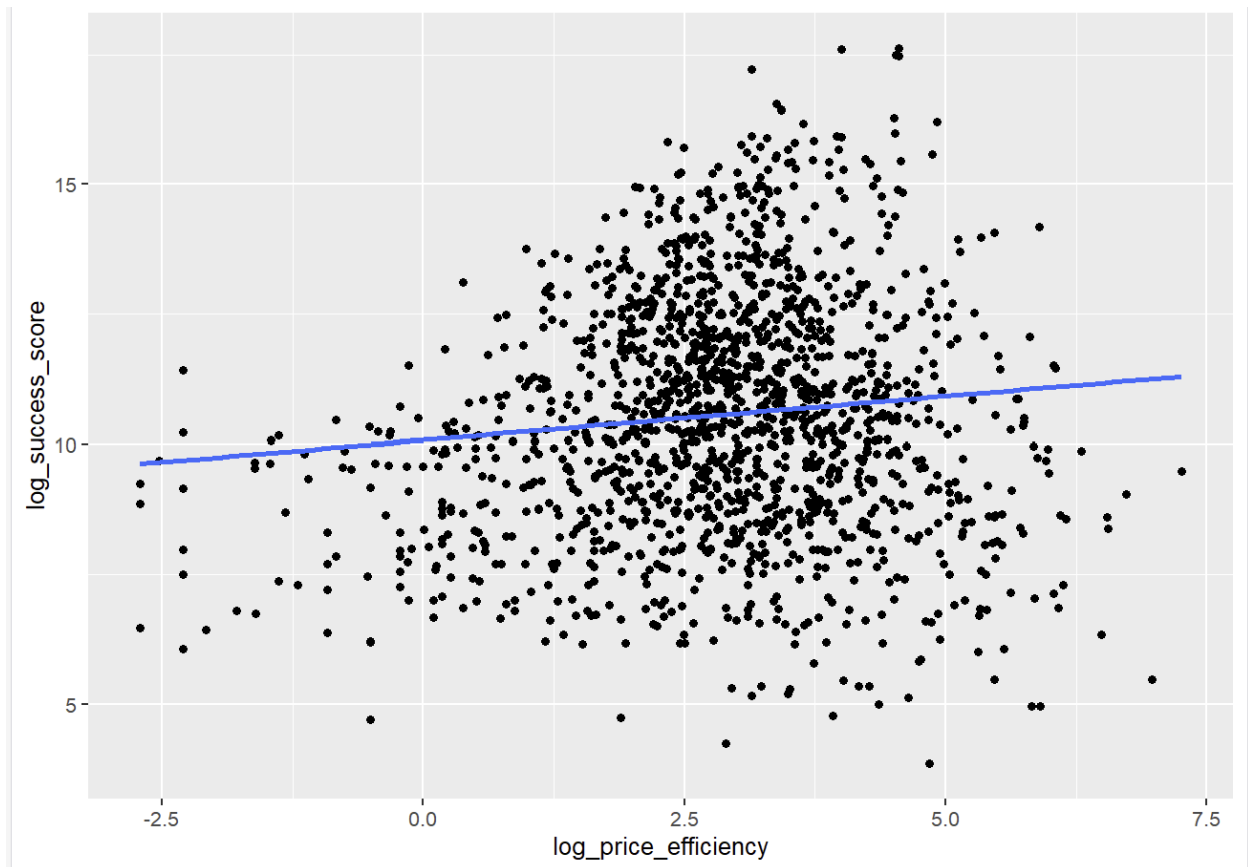
We used a Welch two-sample t-test because we are comparing the mean log success scores of two independent groups. The test results show that the mean log success score for multiplayer games is 11.06134, while the mean log success score for single-player games is 10.29326. The test statistic is $t = 6.1993$, with $df = 1037.1$. The two-sided p-value is $8.173e-10$, so the one-sided p-value for testing whether multiplayer games have higher success is about $4.0865e-10$. This p-value is much smaller than 0.05.

The 95% confidence interval for the difference in mean log success between multiplayer and single-player games is (0.5249629, 1.0112027). Since this interval is entirely positive and does not contain 0, it supports the conclusion that multiplayer games have a higher average log success score than single-player games.

We also checked the model diagnostics. The residuals vs fitted plot shows two vertical groups, which is expected because player mode is a categorical variable with two groups. The Q-Q plot is close to a straight line, so the normality condition is reasonably satisfied. The scale-location plot does not show a major difference in spread between the two groups, and the residuals vs leverage plot does not show serious influential points. Therefore, the conditions for this comparison are reasonably met.

Since the p-value is much smaller than 0.05, we reject the null hypothesis. There is strong statistical evidence that multiplayer games achieve higher success than single-player games. Based on this sample, multiplayer games are more successful on average than single-player games.

3.8 Question 7: Is price efficiency (median hours per dollar) associated with success?



For Question 7, we examined whether price efficiency, measured as median hours per dollar, is associated with success. Since the scatterplot does not show a clear linear pattern and the points are widely spread out, we used a Spearman correlation test to measure the monotonic relationship between log_price_efficiency and log_success_score. The Spearman correlation coefficient was 0.073, which is very close to 0. This indicates a very weak positive association, but the relationship is so small that there is little to no meaningful monotonic relationship between price efficiency

and success. Therefore, based on this sample, games with higher price efficiency do not appear to have substantially higher or lower success scores.

4. Conclusion

In conclusion, our data analysis shows that success on Steam games is shaped by multiple factors rather than one single factor. Metacritic score was a stronger predictor of success compared to Steam user scores, although using both together created the strongest model. This suggests that both critic reviews and user scores matter but they capture different aspects of a game's performance. Genre also played an important role in our experiment. Action, RPG and Simulation games were more associated with success, while indie games had lower success scores after holding other genre tags constant. This suggests that larger, higher budget games tend to have more visibility, prices and more reviews.

We also realized that some of our findings challenged our hypothesis. Metacritic score ended up being a better predictor of success than user score. This might be because Metacritic aggregates pre-purchase signals that drive sales, whereas user review percentages can be high even for niche games with limited commercial reach.

Average playtime was counterintuitively a better predictor of success than median playtime. This might be due to two reasons. First, on a log scale, the heavy upper tail of `average_playtime` is compressed enough that the residuals are reasonably

well-behaved. Second, the very long-tail players whose hundreds or thousands of hours pull the mean up are themselves an informative signal that a game that retains super-fans is, in fact, a more commercially successful game. The mean captures this engagement-tail information that the median, by construction, discards.

Multiplayer games had higher success scores compared to single player games, which could be for several reasons. Multiplayer games have stronger network effects, longer lifecycles, and are likely more correlated with more successful genres such as action.

We also expected price efficiency to predict success on the theory that customers reward developers who deliver more hours per dollar. Instead, the relationship is statistically detectable only because $n = 1,553$ is large, and is substantively negligible.

4.1 Limitations

Several caveats apply. First, our success metric, $\text{success_score} = \text{num_reviews_total} \times \text{price}$, mechanically incorporates price, so the Q4 result on price is partly tautological. Second, the analytic sample is restricted to paid games with a Metacritic score, a population that systematically over-represents larger studios and under-represents the long tail of free-to-play and unrated indie releases. The negative Indie coefficient in Q3 should be read in this context. It describes Indie-labeled games that nonetheless cleared the Metacritic threshold, not the indie scene as a whole. Third, all regressions are linear on the log scale and assume the chosen predictors enter monotonically.

Non-linear or threshold effects (for example, a price ceiling above which sales collapse) cannot be detected by this specification.

Fourth, the data are a single-time-point snapshot from March 2025 and therefore confound a game's age with its accumulated success. A 15-year-old game has had longer to accumulate reviews than a 6-month-old release at any given price. We did not adjust for release date in this analysis.

4.2 Future Directions

Three extensions would strengthen and broaden the findings reported here. First, incorporating release date would let us separate per-year accumulation effects from per-game success and would allow us to distinguish enduring hits from one-week flashes. Second, a more complete outcome metric like peak concurrent users, an estimated-owners midpoint, or revenue inferred from third-party trackers, would relax the dependence on the review-count proxy and would allow a cleaner test of the price hypothesis. Third, a more detailed model. Analyze if the Multiplayer advantage depends on genre. Does the Metacritic effect taper above a threshold?

Practically, for an independent developer choosing between project concepts, this analysis suggests prioritizing a strong Metacritic-eligible release in a Simulation, RPG, or Multiplayer space, pricing for the budget tier rather than for hours-per-dollar value,

and recognizing that the Indie tag itself, irrespective of execution quality, operates as a commercial headwind on Steam.

To sum it up, Steam game success appears to be dependent on factors like critic reception, genre, price, engagement and multiplayer function. Although our sample only included paid games with Metacritic score, user reviews and playtime that is greater than 0. Our finding doesn't reflect all the Steam games, especially free to play games.